

recruitment-visualization-process

Data wrangling and cleaning

Several modifications are required for the dynamic coral data prior to functional analysis. The following coding block details handling missing data and establishing a shared color gradient for each taxa.

```
library(tidyverse)
library(ggthemes)
clean_coral <- read_csv(here::here("vedika-r-files", "data", "coral_tidy_2024_with_dynamic_colorway.csv"))

# Coral color way - consistent across all visualizations
coral_colors <- c(
  Acr = "blue",
  Poc = "red",
  Por = "darkgreen",
  Mil = "darkorange"
)

clean_coral <- clean_coral |>
  # Removing plots from backreef surveys
  filter(str_detect(transect, "T")) |>
  # Change volume variables to numeric
  mutate(
    height = as.numeric(height),
    width = as.numeric(width),
    length = as.numeric(length),
    # Create volume (cm^2), z (m), and y (m) columns
    y_m = y / 100,
    z_m = z / 100,
    # Acropora were input as "A" in 2024, converting to "Acr"
    taxa = case_when(taxa == "A" ~ "Acr",
```

```
TRUE ~ taxa))
```

Plotting Recruitment Overtime at Transects

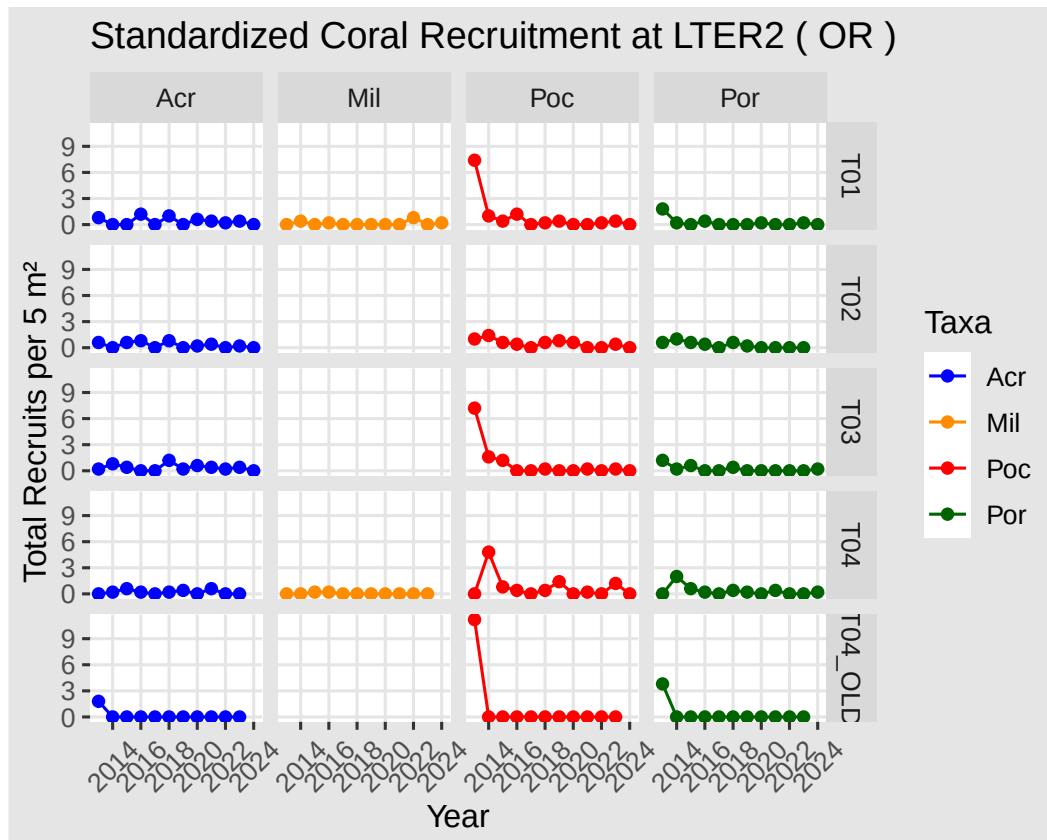
The function `transect-recruitment` is sourced locally from the `recruitment-visual-function.R` file stored in this repository. This function is intended to create the total standardized recruitment per year at each transect for a specified site. Each figure generated is stored in its respective subfolder (`transect-site-recruitment`).

```
source(here::here("vedika-r-files", "recruitment-visual-function.R"))
```

```
# Generate plots for each transect (ex. at site LTER1 for BR)
```

```
transect_recruitment(coral_data = clean_coral, site_name = "LTER2", habitat = "OR")
```

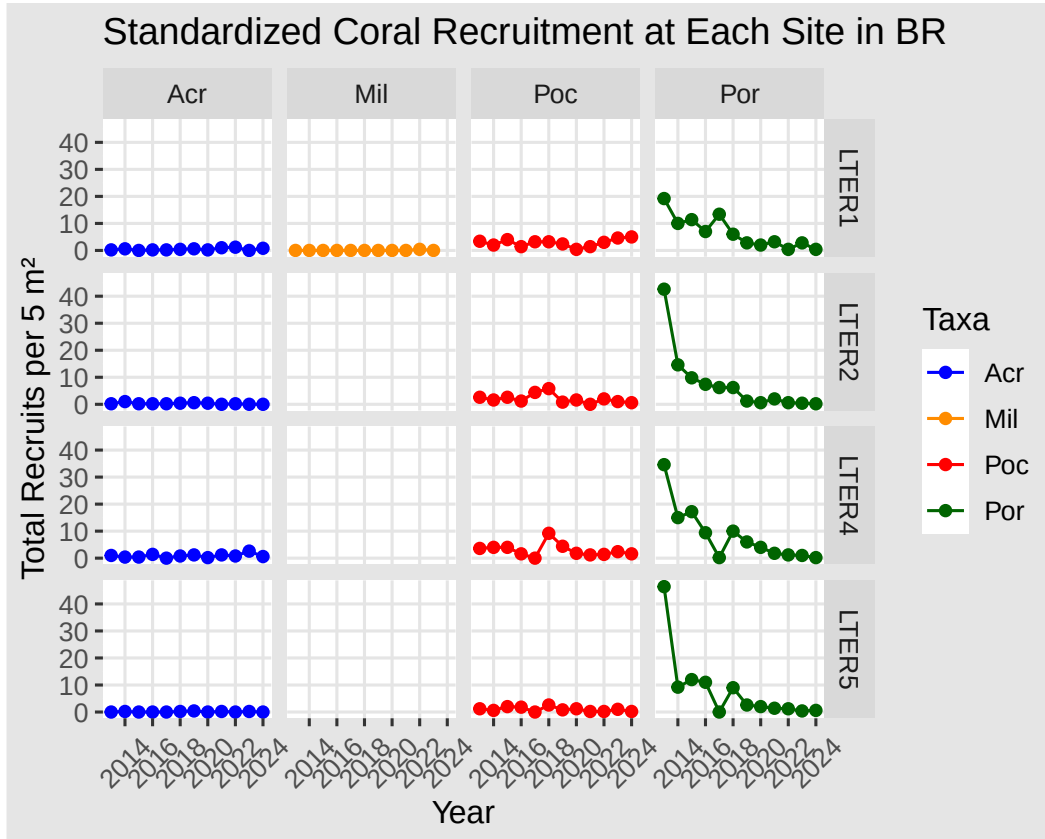
```
[1] "T01"      "T02"      "T03"      "T04"      "T04_OLD"
```



Plotting Recruitment Overtime at Sites

The function `site-recruitment` is sourced locally from the `recruitment-visual-function.R` file stored in this repository. This function is intended to create the total standardized recruitment per year at each site for a specified habitat across all taxa. Each figure generated will be stored in its respective subfolder (`transect-site-recruitment`).

```
# Generate plots for each site at a specific habtat
site_recruitment(coral_data = clean_coral, habitat_name = "BR")
```



Plotting Recruitment Overtime at Each Habitat

The function `habitat-recruitment` is sourced locally from the `recruitment-visual-function.R` file stored in this repository. This function is intended to create the total standardized recruitment per year at each habitat across all taxa.

```
# Generate plots for each habitat
habitat_recruitment(clean_coral)
```

